

Modeling talent development pathways in alpine ski racers

Rhiannon L. Cowan^{a,*}, Bradley Fawver^{a,b}, Keith R. Lohse^a, Troy Taylor^c, Paul R. Ford^d, A. Mark Williams^a

^a University of Utah, College of Health, Department of Health and Kinesiology, 383 Colorow Drive, Suite 260, Salt Lake City, UT, 84108, USA

^b US Army Medical Research Directorate-West, Walter Reed Army Institute of Research, USA

^c U. S. Ski and Snowboard, 1 Victory Lane, Park City, UT, 84060, USA

^d St Mary's University, Waldegrave Rd, Twickenham, TW1 4SX, UK

ARTICLE INFO

Keywords:

Expertise

Practice

Engagement

Mixed-effects models

ABSTRACT

We explore developmental pathways to excellence in sub-elite adolescent alpine ski racing by exploring athletes' engagement in practice activities that characterize sport specialization and/or diversification. Moreover, we extend previous research by using more refined linear mixed-effects models to examine changes in practice and performance data over time rather than relying on the less sensitive repeated measures ANOVAs typically used in previous research. Altogether, 174 individuals ($M = 15.9$ yrs, women = 91) from professional development academies in the United States completed a sport-specific Practice History Questionnaire (PHQ). We recorded national rankings in technical events to assess performance. Mixed-effects models indicated that hours in ski-related activities generally increased over time (linear $\beta = 1.38$, p 's < 0.001), except time spent in play, which decreased over time ($\beta = 102.8$, $p = 0.002$). Performance rankings also improved over time ($\beta = 30.99$, $p < 0.001$), with more time spent in ski-activity associated with greater rates of improvement, particularly in younger skiers ($\beta = 9.77$, p 's < 0.001). The effects were largely driven by time spent in coach-led group practice ($\beta = 2.16$, $p < 0.001$). Overall, an earlier age of first engagement was associated with better rankings ($\beta = -15.40$, $p = 0.024$), and accumulating more hours in other sports was also associated with greater rates of performance improvement ($\beta = -8.87$, $p = 0.014$). Sub-elite alpine ski racers engage early in the sport, with higher-ranked athletes spending more time in ski-specific practice, particularly group practice, during development. However, these athletes generally continued to participate in other sports across development, providing greater support for the early engagement hypothesis rather than early specialization per se.

1. Introduction

The development of expertise in sport has long been characterized by two primary developmental trajectories, namely, the specialization and diversification pathways (Coutinho, Mesquita, & Fonseca, 2016). In order to characterize the optimal pathways to excellence across sports, it is necessary to record sport-specific and non-sport specific practice activities that athletes engage in at different stages of development. While the specialization and diversification pathways are frequently presented as being dichotomous, the two pathways may more accurately be viewed as the polar ends of a continuum. Moreover, the early engagement hypothesis has been proposed more recently, which embraces both pathways simultaneously (Ford, Ward, Hodges, & Mark Williams, 2009). The relative importance of specialization, diversification, and early engagement likely fluctuates across sports, thereby impacting on

athletes' progression to elite levels in unique, domain-specific ways.

In this paper, we explore the presence of these different pathways in the sport of alpine ski racing. Although ski racing is often viewed as the marquee sport of the Winter Olympics, no previous researchers have examined the prevalence of these different pathways to excellence in the domain. In addition to extending theoretical knowledge, with a relatively novel population sample, we employ regression analyses to model change over time, which allows for the relative importance of specialization, diversification, and engagement to be examined throughout the development pathway. Previously, researchers have generally used repeated measures ANOVA and simple correlation analyses to study these developmental pathways. The use of regression analyses to model change over time presents a relatively novel approach, allowing us to directly link engagement in different practice activities with changes in objective performance markers. In sum, the use of linear models and

* Corresponding author. 383 Colorow Drive, Suite 260, Salt Lake City, UT, 84108, USA.

E-mail address: rhiannon.cowan@utah.edu (R.L. Cowan).

<https://doi.org/10.1016/j.psychsport.2021.101942>

Received 24 April 2020; Received in revised form 4 February 2021; Accepted 26 March 2021

Available online 8 May 2021

1469-0292/© 2021 Elsevier Ltd. All rights reserved.

linear mixed-effects regressions to analyze patterns of engagement and objective performance over time presents a more refined methodological and analytical approach than previously employed (for exceptions, see Hodges, Starkes, Nananidou, Kerr, & Weir, 2004; Orr, 2007).

1.1. Pathways to expertise

The process of developing athletic expertise is complex and influenced by numerous factors. As a result, researchers seeking to understand why some athletes eventually develop into elite-level competitors, while others fail to reach the upper echelons of their sport, often measure a number of variables, including: (a) the age at which individuals engage in certain activities such as formal practice and competition (Ford, Ward, et al., 2009); (b) the different types of activities athletes engage in at certain ages such as practice, play, and indirect involvement (Cote, Baker, & Abernethy, 2003, pp. 89–110); and (c) the degree to which individuals specialize or diversify in different sports across childhood and adolescence (Côté, Baker, & Abernethy, 2007, pp. 184–202). In this vein, scientists have proposed different pathways to explain the best combination of timing, involvement, and specialization (or diversification) for superior developmental trajectories (e.g., Deliberate Practice Framework, Developmental Model of Sport Participation; Ericsson, Krampe, & Tesch-Römer, 1993; Cote et al., 2003, pp. 89–110). In order to provide some context regarding existing measures and pathways, we briefly review previous frameworks and approaches to quantifying factors associated with the development of expertise.

The traditional view across numerous sport and non-sport domains is that accumulating thousands of hours of practice is critical to the development of expertise (Ericsson, 2006, pp. 685–706; Ericsson & Smith, 1991; Ericsson, Whyte, Ward, Goulding, & Williams, 2017). For instance, a positive relationship has been reported between practice time and performance across various sports and athlete populations (e.g., Baker, Côté, & Deakin, 2005; Baker & Young, 2014; Ericsson, 2020; Helsen, Starkes, & Hodges, 1998; Law, Côté, & Ericsson, 2007; Starkes, Deakin, Allard, Hodges, & Hayes, 1996, pp. 81–106). Published literature suggests that engaging in appropriate practice activities is particularly relevant during childhood and adolescence, when aspiring elite athletes tend to separate themselves from their comparatively less-elite peers (Ford & Williams, 2017, pp. 117–132; Ford, Williams, & Hodges, 2009; O'Connor, Larkin, & Williams, 2016; Roca, Williams, & Ford, 2012; Ward, Hodges, Starkes, & Williams, 2007). The deliberate practice framework, initially proposed over 20 years ago by Ericsson et al. (1993) emphasizes the importance of athletes engaging in effortful, structured activity with the purpose of improving performance (see Baker & Young, 2014). Several predictions proposed by deliberate practice theory are often examined, such as the degree to which practice needs to be 'deliberate', the necessity of reaching certain conventional practice hour milestones such as 10,000 h, and the reduced emphasis on competition time. More recently, the deliberate practice theory has been re-evaluated, with an emphasis on *structured practice*, and how hours in practice alone cannot predict the acquisition of expertise (Ericsson & Harwell, 2019; Macnamara, Hambrick, & Oswald, 2014).

Some researchers have suggested that the age at which individuals begin sport-specific training, as well as achieving other important milestones (e.g., first regional/national competition), has a significant impact on development. Ford and Williams (2012) reported that academy-level football (soccer) players in England, who progressed to the professional level at 16 years of age, reported a higher volume of practice hours across development and achieved sport-specific milestones at an earlier age compared to those who did not reach the professional level. The authors proposed an 'early engagement' hypothesis, arguing that in some sports, notably football, children are typically exposed to the sport at an early age (i.e., 5–10 years old) and consequently, they tend to accumulate a higher volume of sport-specific training hours, including increased participation in competitive leagues and tournaments, compared to those who specialize later (Baker,

Cobley, & Fraser-Thomas, 2009; Côté, Lidor, & Hackfort, 2009; Coutinho, Mesquita, Davids, Fonseca, & Côté, 2016). Other important milestones for development include the ages at which children first engage in practice with a coach, structured competition, or non-sport-specific training (e.g., resistance/weight training, plyometrics; Walters, Read, & Estes, 2018). Moreover, Ford, Williams, and Hodges (2009) reported that soccer players who engaged in a greater volume of unstructured 'play' hours during development typically reached higher skill levels than players who engaged in less play. It seems that unstructured activities may help autonomous skill development, particularly where athletes may be over-exposed to coaching that is too structured and prescriptive (Ford & Williams, 2012). Therefore, early engagement in play may be beneficial to the development of expertise in multiple domains. While stories of 'late bloomers' exist across some sports, athletes who first engage in their primary sport later in life might experience a 'practice deficit', accumulating relatively fewer hours of practice and play in the sport compared to those who first engage in practice and competition at a younger age. Yet, for athletes transferring to alpine ski racing from another winter sport, or between alpine ski racing disciplines (i.e., speed to technical events) later in their career, talent and skill transfer may be possible due to common foundational characteristics between athletic domains (i.e., similar co-ordination, balance, and kinesthetic factors between sports/disciplines).

The idea that specializing yields some long-term benefits for aspiring elite athletes (e.g., greater accumulation of domain-specific practice hours across an individual's career) has become a cornerstone of sport culture. Furthermore, the argument that specialization is seemingly more pertinent for placement in sub-elite and elite level academies and clubs (e.g., AAU teams), as well as collegiate scholarships, has increased in recent years. However, the early engagement hypothesis does not propose early specialization *per se*, because engagement in other sports could be beneficial for growth and development, particularly when it does not come at the expense of accumulating practice time in the target sport for expertise. For instance, even among athletes who first engage in a sport from an early age (e.g., 2–4 years old), many continue to participate in other sports to varying degrees alongside the target sport, termed sport diversification. It has been suggested that diversifying across sports during childhood and adolescence may precede the development of expertise in adulthood, whereas early specialisation might be more predictive of adolescent success (e.g., DiFiori et al., 2018). Diversification is thought to affect development of expertise through the transfer of physical attributes, psychological resiliency, and technical/tactical skills to a primary sport later in development (Abernethy, Baker, & Côté, 2005; Coutinho, Mesquita, & Fonseca, 2016). Many elite competitors anecdotally attribute part of their success to other sport experiences during childhood and adolescence that facilitated creativity, novel problem-solving skills, and unique motor skills that subsequently became valuable for the primary sport later in development. Moesch, Elbe, Hauge, and Wikman (2011) reported that many elite athletes train less during childhood as a result of career planning and only specialize in their primary sport later in development. Furthermore, the authors suggest that the number of other sports that athletes engaged in during childhood is less important than the intensity of practice during adolescence, which tends to be greater in elite athletes compared to their peers.

1.2. Specialization versus diversification

Although proponents of either specialization or diversification propose contrasting pathways to excellence (see Côté et al., 2007, pp. 184–202; Ford & Williams, 2017, pp. 117–132), researchers have argued that these developmental routes are not dichotomous (see Abbott & Collins, 2004; Côté & Vierimaa, 2014). A growing belief is that early engagement and diversification can co-exist at critical moments during development (Jayanthi, Pinkham, Dugas, Patrick, & LaBella, 2013). The

Developmental Model of Sport Participation (DMSP) details different stages of development in which athletes might diversify and engage in more play activities during youth (i.e., sampling phase) and only specialize and invest in high performance later in life (Cote et al., 2003, pp. 89–110). Such an integrated developmental trajectory could particularly suit sport domains with unique seasonal constraints that could therefore encourage diversification, such as alpine ski racing. However, to date, few researchers have examined developmental pathways in alpine ski racing, including the degree to which these athletes specialize or diversify, engage in serious training early or later, or engage in different sporting activities during development. Those who have conducted research in this area have focused more on the psychological characteristics of alpine ski racers, rather than associations between practice and performance (e.g., Fawver et al., 2019).

1.3. Modeling developmental pathways to expertise

In order to model developmental trajectories longitudinally, without actually tracking athletes for years, researchers have relied on retrospective practice history profiling to explore factors believed to influence development, such as the age of first engagement in the sport, sport-specific milestones, and types of practice activities (e.g., Ford, Low, McRobert, & Williams, 2010; Ford & Williams, 2012). These data are typically analyzed using repeated measures analyses of (co)variance to compare group differences over time (Law et al., 2007; Wall & Côté, 2007). However, retrospective (and prospective) sampling of data longitudinally often leads to missing observations because athletes can abstain from sport participation for a period of time, miss practice or competition due to injury, or fail to log hours for a particular season. The statistical approaches most typically used by researchers cannot appropriately account for these missing observations without listwise removal/deletion of participant data, thereby reducing statistical power (Garcia & Marder, 2017), or imputing missing responses with estimated (i.e., mean, median) data. Furthermore, the assumption of equal variance, which is central to ANOVA/MANOVA, may not be satisfied because practice and performance changes over time, and in some cases, would be violated by the nature of the sample and measures being employed.

In contrast, mixed-effects regression is more adept at handling missing data and changes in variance over time due to the lack of assumptions regarding the independence of observations (Bates, Mächler, Zurich, Bolker, & Walker, 2015; Garcia & Marder, 2017). For example, linear mixed-effects regression allows researchers to dynamically handle missing data and reveal patterns between variables over time. Moreover, mixed-effects regression approaches provide additional information on average group trends and individual performance changes by examining changes in slope trajectories over time. The use of such innovative methods has the potential to yield important insights into how small variations in practice type and volume are related to performance.

Hodges et al. (2004) were the first to employ hierarchical regression modeling to study the development of expertise in an athlete sample using measures of engagement in sport-specific and non-sport-specific training, as well as early career milestones. Using linear trends across time and simple correlations between variables and performance in a sample of triathletes and swimmers, this novel work advanced our understanding of the contributions of different training activities for development in sport. Hodges et al. (2004) reported that triathletes needed more variety in practice activities to attain expertise, while swimmers required more sport-specific practice. The authors conclude that this difference was due to the sport focus and constraints within each domain. General physical fitness-related activity did not predict performance times in triathletes, but it did in swimmers, aligning with Ericsson et al. (1993) who posited that engaging in physical 'maintenance' activity is important for overall fitness, while not necessarily being indicative of superior performance trends. While this approach was innovative, the statistical techniques applied (e.g., group

differences, correlations, simple regression) did not capture important information related to changes within individuals over time, which can further improve our understanding of the talent development processes. In this study, we apply more complex statistical techniques to analyze objective performance markers and practice history data. Mixed-effects models allow developmental trajectories to be plotted over time, while showing the effect of independent variables on the slope of the trajectory (i.e., volume of hours in ski-related practice).

1.4. The current study

In the present study, we examine the extent to which sub-elite alpine ski racers engage in specialization, diversification or early engagement pathways. Specifically, we investigate the relative importance of sporting milestones and different types of engagement activities for performance outcomes and the development of expertise in a domain with comparatively limited research, namely, alpine ski racing. We extend the methodological approach employed by Hodges et al. (2004) and Fawver et al. (2019) by using mixed-effects regressions to explore within- and between-participant variability in practice and performance over time.

A large sample of alpine ski racers enrolled at developmental academies across the U.S. completed an adapted, ski-specific version of the Practice History Questionnaire (PHQ; Ford, Williams, & Yates, 2010), which included a variety of sport-specific milestones, type and hours of practice in each year the athlete participated in ski racing, and time spent in other sport activities during development. Additionally, national points (i.e., ranking) in the technical disciplines (i.e., slalom, giant slalom) were collected for each participant in every year they were available. The following hypotheses are somewhat speculative and based on related research from other domains, giving a more detailed approach to practice and performance over time than previous reports on the developmental history profiles of elite athletes in winter sports (Fawver, 2019). First, we predicted that practice hours would increase over time (i.e., years) in conjunction with performance ranking (Baker, 2003; Moesch et al., 2011). Second, fitting with the broad literature base, we expected that alpine ski racers with a lower ranking (i.e., better performance) would have engaged in significantly more practice hours compared to individuals with a higher ranking (Baker et al., 2009; Hodges et al., 2004; Law et al., 2007). Third, we predicted that athletes who reported first engaging in practice, competition, and non-ski training at an earlier age would report more practice hours across their careers and demonstrate a superior ranking to those who achieved milestones at an older relative age (Baker et al., 2009; Coutinho, Mesquita, & Fonseca, 2016; Ford & Williams, 2012). Fourth, we forwarded no directional hypotheses related to the time spent in different types of ski-related activities (i.e., coach led group, coach led individual, individual, play, indirect); however, we speculate that, other than traditional modes of practice/exposure, those athletes reporting more individual, play, and indirect hours would have a superior performance trajectory compared to their peers who engaged in less unstructured activity. Finally, we offered no prediction related to whether alpine ski racers engaged in a specialization, diversification, or the early engagement pathway, although anecdotal evidence suggests the domain is characterized by early first engagement (~2 years old) as well as ski racers engaging in sports during the off-season.

2. Methods

2.1. Participants

Altogether, 174 youth alpine ski racers (women = 91) from five different professional development academies across the U.S. volunteered to participate. We have previously published data from this sample in Fawver et al. (2019), but the current paper uses a subset of the data to address a series of novel questions. Participants (*M* age = 15.9

years, $SD \pm 1.3$, Range = 13–19) were born between May 31st, 1998 and August 15th, 2008. All academies were current members of the United States Olympic Committee High Performance Center program. Participants competed in one or more alpine ski racing disciplines. Additionally, participants had current rankings at the National level and, if old enough, Fédération International de Ski (FIS) points (subset $n = 66$). After receiving approval from the University Institutional Review Board, written parental consent and assent were obtained for participants prior to data collection.

2.2. Measures

Practice History Questionnaire (PHQ). We used an adapted version of the PHQ by Ford, Low, et al. (2010) to collect demographic information and record the ski- and sport-related activities across development for each participant. The only change to the original PHQ was the use of ski terminology to make the questionnaire sport-specific. The first section of the PHQ focused on general and ski-specific milestones. Participants noted the age at which they first began skiing, age they first participated in supervised skiing, age of first practice/training and competition, and various start ages of participation into different competition levels (e.g., junior state championships). The second section of the PHQ asked participants to give details about their engagement in ski-related activities from the age they first began the sport. The activities were divided into six categories of engagement: (a) competition; (b) one-to-one coach-led practice; (c) group coach-led practice; (d) individual practice; (e) free play; and (f) indirect involvement. Participants were asked to record the number of months a year that they spent in each activity and the number of hours spent in the activity each week. Participants noted the number of weeks from each year that they were injured and unable to take part in ski-related activities. The final section asked participants to list their engagement in other sporting activities that they had taken part in for longer than three months, including start age, number of hours spent per week and months per year, and the standard (i.e., quality or level of competition) of participation.

Performance. Alpine ski racing National and FIS International points are calculated based on race times and results continuously throughout the competitive season. The United Ski and Snowboard Association (U.S.S.A.) database provided current rank and points for each athlete who participated across the four main alpine ski racing disciplines (downhill, slalom, giant slalom, and super-G slalom). Given the amount of performance data (i.e., rankings) available, as well as limited focus on speed discipline-specific training in the academies (technical events = 399 observations (rankings across all years) from 160 participants; speed events = 291 observations for 140 participants), we chose to only analyze performance in the technical discipline (slalom and giant slalom). The highest ranking for each participant in either slalom or giant slalom for each year was used as the technical rank. Note that when skiers first begin competing on the national circuit, they have a performance score of 999.9, and this ranking improves at a faster rate in the first few competition seasons as ski racers quickly reach their skill level within the standings. When the ski racers are regularly competing against peers of similar skill level, it is harder to improve ranking at a fast rate. Furthermore, the degree to which a ski racer can improve is relatively less when ranked higher (e.g., it is harder to move from a ranking of 200 to 190 than from 999.9 to 989.9).

Procedure. Two researchers visited each academy to supervise data collection, with questionnaires completed via pen and paper within a ~90-min workshop. Verbal instructions detailing the structure and answer formatting of each section of the PHQ were given (i.e., a detailed explanation was given for the hours of practice section, which included responses for monthly, yearly, and hourly time spent in a variety of ski-related and ski-unrelated activities). Participants were asked not to talk or view other questionnaires during this time or consult with their coaches. The workshop ran until all participants had completed the PHQ. A total of 17 participants were removed from the analysis because

of errors in filling out the questionnaires.

2.3. Statistical analyses

All analyses were conducted using R 3.4.1 (R Core Team, 2017) and R Studio 1.1.463 (Boston, MA; Bates et al., 2014; Kuznetsova, Brockhoff, & Christensen, 2017). Full details of all statistical models are provided in the supplemental appendix. The R packages used for analysis can be found in the following citations (Bates et al., 2015; Kuznetsova et al., 2017; Lüdtke, 2020). Our models focused on either time in practice or national ski points in the technical disciplines as the dependent measure. For each of these outcomes, there were different amounts of data available. For instance, participants had more years with reported hours of training than years with national technical points. To address these differences, we started with unconditional models for the effect of time. Model fit was assessed using the Akaike Information Criterion (AIC), and we progressively added polynomials (linear, quadratic, and cubic model comparisons) for the curvilinear effect of time until we arrived at the best fitting model. The best fitting model was based on the random effect of time, consequently, we kept effects that led to a reduction in the AIC for the dependent variable. Cubic and quadratic comparisons were used to find the best fit of the fixed effects moving forward (Long, 2012). Subsequently, we added all of the fixed-effects of interest and their interactions into each model, based on A-Priori hypotheses (all models presented in [supplementary material Table 6.a - 6.s](#)). Random intercepts and slopes were included in all models for the linear effect of time. We also calculated an r^2 approximation to estimate the variance explained by each model's fixed effects, referred to as a pseudo- r^2 (Byrnes, 2008). Given the differences in each of these models, we provide a brief description of the model preceding a description of the results below. In each model, the statistical significance of individual parameters was assessed with a t -test using degrees of freedom based on the Welch-Satterthwaite approximation. The significance level for all statistical tests was set at $\alpha = 0.05$.

3. Results

3.1. Descriptive statistics and milestones

We present the descriptive statistics for performance, milestones, and ski engagement in [Table 1](#). The ski racers first engaged in the sport at an early age ($M = 2.7$ years of age, $SD \pm 1.53$) and began supervised ski training a few years later ($M = 6.17 \pm 2.02$ yrs). Similarly, ski racers first competed in a ski race relatively early ($M = 7.45 \pm 1.75$ yrs) and began non-ski specific training prior to adolescence ($M = 11.02 \pm 2.54$ yrs). When looking across their developmental spans, participants accumulated a mean of 489.85 h of practice per year in ski racing, not including indirect hours, and engaged in five other sports across development ($M = 5.26$ other sports, $SD \pm 2.21$).

3.2. Hours in practice over time

Types of Activities. As shown in [Fig. 1A](#), hours in practice generally increased (non-linearly) across time, with a greater rate of increased hours in later years ($\beta = 1.38$, $p < 0.001$), quadratic ($\beta = 0.099$, $p < 0.001$), and cubic ($\beta = 0.003$, $p < 0.001$). This increase was true for all types of activities (p 's < 0.001) except for play, which actually decreased over time ($p = 0.002$) (see [Fig. 1B](#)). The full statistical analysis of hours in practice over time is reported in the supplemental appendix.

Developmental History. To examine how ski racers' developmental histories affected hours in practice over time, models focused on total hours of ski-related activities (except for indirect hours) to reduce the potential number of statistical tests. Furthermore, practice history was separated into two different categories of variables: (a) the age of first engagement in a number of ski-related activities (i.e., ages of first ever engagement with skiing, supervised training, competition, and non-ski

Table 1
Descriptive statistics (means and standard deviations) for the sample of academy alpine ski racers.

Measure	Subscale	Men	Women	Combined	N
		M (SD)	M (SD)	M (SD)	
Age	Age in 2018 (y)	16.00 (1.36)	15.70 (1.30)	15.82 (1.34)	169
Performance	National	91.47 (39.98)	116.51 (54.96)	105.98 (50.00)	133
	Technical				
Average hours of engagement per year	Points (Rank)				
	Total Hours (h/y)	463.47 (299.80)	515.21 (330.74)	489.85 (315.87)	161
	Competition (h/y)	108.07 (124.92)	107.17 (80.16)	107.97 (102.59)	161
	Coach-led 1-on-1 practice (h/y)	57.16 (85.65)	94.72 (136.22)	77.80 (117.08)	161
	Coach-led group practice (h/y)	160.39 (148.32)	179.52 (161.03)	168.5 (155.65)	161
	Individual practice (h/y)	50.95 (80.70)	56.86 (101.82)	54.31 (92.31)	161
	Play (h/y)	89.46 (75.94)	76.95 (58.18)	82.46 (66.85)	161
	Indirect (h/y)	145.93 (271.26)	141.34 (225.48)	141.98 (246.59)	161
	Total Hours	323.61 (420.15)	364.84 (412.94)	338.95 (398.68)	158
	Other Sports (h/y)				
Sport-specific milestones	First engagement (y)	2.57 (1.28)	2.84 (1.74)	2.71 (1.53)	166
	First supervision (y)	5.86 (2.08)	6.42 (2.04)	6.17 (2.02)	167
	First competition (y)	7.30 (1.63)	7.58 (2.02)	7.45 (1.75)	169
	First non-ski training (y)	11.44 (2.74)	10.62 (2.61)	11.02 (2.54)	168
	Total	4.97 (2.29)	5.58 (2.14)	5.26 (2.21)	169
Number of other sports	2–5 years old	1.40 (1.50)	1.88 (1.64)	1.64 (1.58)	168
	6–12 years old	4.22 (2.07)	4.86 (2.22)	4.52 (2.15)	168
	13–18 years old	3.59 (1.95)	3.30 (1.63)	3.43 (1.77)	167

Note. N = number of participants with data available for analysis; y = years old; h/y = hours per year.

training); and (b) the total number of hours played in other sports (reflecting relative ‘specialization’).

Participants first started skiing at a very young age (median = 2 yrs),

with age of first supervision (median = 6 yrs) and first competition (median = 7 yrs) following a few years later (Fig. 2A). The only milestone to regularly occur during/near puberty was first engagement in non-ski training (median = 12 yrs). Altogether, 158 ski racers in the sample participated in multiple sports (Fig. 2B); however, the number of sports with which they engaged changed over time.

To assess the effect of achieving different milestones on time spent in practice, we focused on the relationship between total hours of practice and age of first engagement in skiing. We created a variable that coded whether the age of engagement was less than or equal to the median age of 2 years, or greater than the median age. We added this variable Age of Engagement to a model with Time (linear, quadratic, cubic), Age in 2018, and all interactions (see supplemental appendix for full model details). No statistically significant effects of Age of Engagement ($p = 0.542$) or its higher-order interactions (p 's > 0.094) emerged in these models. Although participants who started ski racing earlier reported greater average time in practice and a steeper trajectory for hours per year (see Fig. 1C), there was not a statistically discernible effect of earlier engagement on total hours of practice across development.

We examined the relationship between total time spent in ski racing activities (excluding indirect hours) and total hours of engagement in other sports across development (see Fig. 2D). Given the high number of practice hours in ski racing, as well as total hours in other sports reported, individuals who reported more than 20,000 h over the course of their development were excluded on the grounds that the data were not reliable (i.e., equating to 6 h of other sports practice every day for ten years). This led to the exclusion of 3.2% of total observations. Within the remaining data, we created a variable of Other Sports that coded if a ski racer accumulated greater than or equal to the median of 3,300 h of total sports throughout their development, or less than 3,300 h. We added the variable of Other Sports to a model with effects of Time (linear, quadratic, and cubic), Age in 2018, and their interactions. After controlling for other factors in the model, there was a significant main-effect of hours in Other Sports on time spent in ski-related activities ($\beta = 1.92$, $p = 0.005$). Overall, there were statistically significant effects of the linear ($\beta = 1.38$, $p < 0.001$), quadratic ($\beta = 0.117$, $p < 0.001$), and cubic ($\beta = 0.004$, $p < 0.001$) effects of time. While there was not a significant effect of Age ($\beta = 2.39$, $p = 0.521$), there were significant interactions of Age x Time ($\beta = -0.199$, $p = 0.035$), Age x Time² ($\beta = -0.04$, $p = 0.01$), and Age x Time³ ($\beta = -0.002$, $p = 0.016$). These findings suggest that ski racers continue to accumulate a high volume of hours in other sports, parallel to accumulating high volumes of hours in ski-related activities each year throughout development. These interactions indicate that accumulation of practice time in skiing is more pronounced for younger ski racers who engage in other sports earlier in development (for details of all parameters, see supplemental appendix).

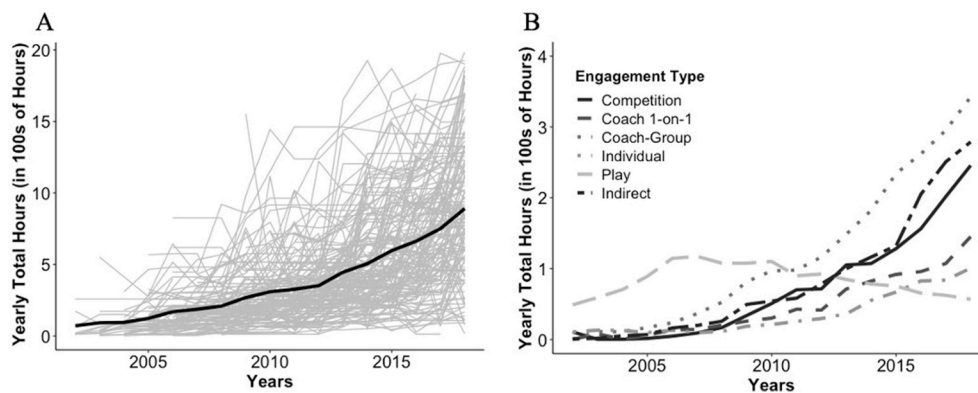


Fig. 1. (A) Total hours of time in ski-related activities per year as a function of time. Thin grey lines show individual participant data, while the thick black line shows the average time in practice across participants (Marginal $r^2 = 0.366$). (B) Time spent in specific ski-related activities per year, with line color and type denoting average time spent in the six different types of activities measured across participants (Marginal $r^2 = 0.187$).

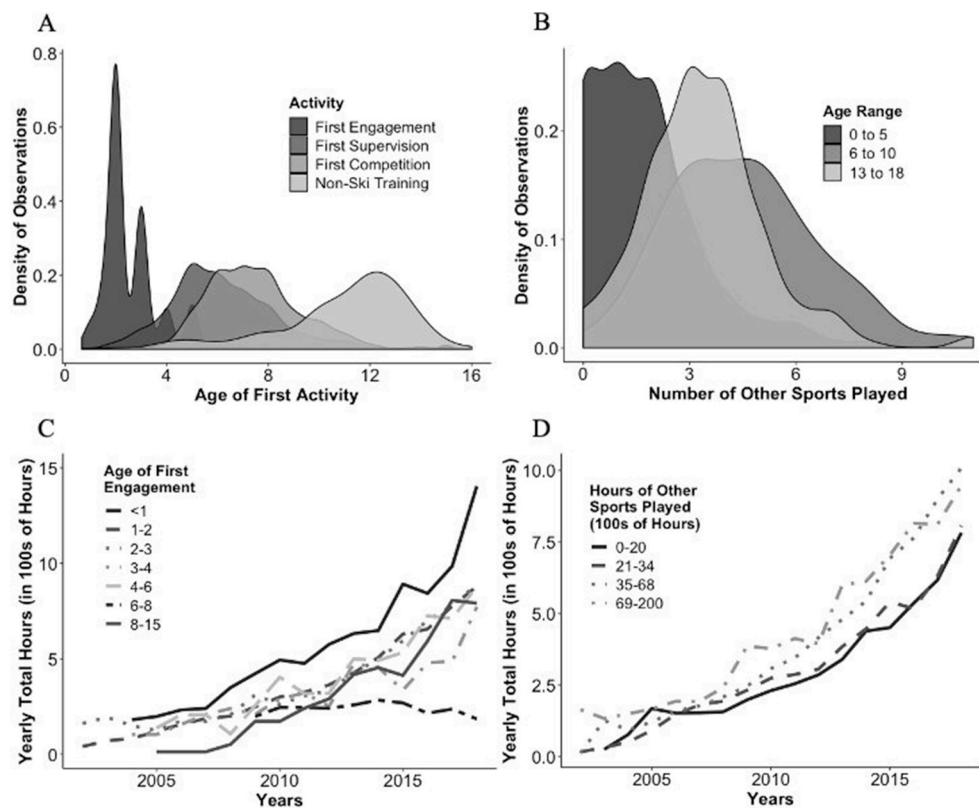


Fig. 2. (A) A density plot showing the relative frequency of different ages for ski racer's developmental milestones. (B) A density plot showing the relative frequency of the number of sports ski racers reported playing in different age ranges. (C) Aggregated data showing the relationship between total hours of practice per year (averaged across participants) based on age of first engagement in skiing (Marginal $r^2 = 0.370$). (D) Aggregated data showing the relationship between total hours of practice per year (averaged across participants) and total hours in other sports played in the ski racer's history (Marginal $r^2 = 0.391$).

3.3. Technical performance over time

Our analysis of national technical points over time showed that participants generally improved their rankings over years. Using a similar curvilinear model building procedure as used for hours, the best fitting models for performance included linear fixed-effects and random-effects of Time. Additionally, inspecting the residuals for models of performance revealed large violations of normality and homoscedasticity. To address this problem, we truncated technical points at 300, excluding the very high point values (6% of observations). These exclusions create some limitations, discussed later, but importantly this led to a normal and homoscedastic distribution of residuals.

Performance as a Function of Time in Practice. For technical performance, national points were regressed onto Time, Age in 2018, and average Hours in different types of activities over a ski racers'

development. We used different models for each type of activity, but all models controlled for Time and Age as covariates (see Table 2). After controlling for time spent in different types of ski-related activities, significant effects of Time, Age, and Time $\times$ Age interactions emerged. Performance in technical events generally improved over time, but the rate of improvement was dependent on an individual's age, with younger ski racers progressing at a faster rate. The effects for hours spent in different activities were less conclusive.

For total hours of ski-related activities (not including indirect hours), there was no main-effect of Hours ($\beta = -0.74, p = 0.375$), but there were significant interactions of Time $\times$ Hours ($\beta = -1.26, p = 0.003$) and Time $\times$ Hours $\times$ Age ($\beta = 0.86, p = 0.006$). Given the significant effect of Time ($\beta = -30.99, p < 0.001$) and the Time $\times$ Age interaction ($\beta = 9.77, p < 0.001$), these effects indicate that individuals who engaged in more total hours of practice in development improved their ranking at a faster rate,

Table 2

Fixed-effects from models of technical performance regressed onto different types of practice hours.

	Practice Type													
	Total Hours		Competition		One-on-One		Coach-Group		Individual		Play		Indirect	
Fixed-Effect	β	SE	β	SE	β	SE	β	SE	β	SE	β	SE	β	SE
Intercept	102.71*	3.39	102.52*	3.38	103.01*	3.37	104.30*	3.41	102.64*	3.38	102.80*	3.41	103.29*	3.43
Time	-30.99*	1.68	-31.04*	1.73	-30.76*	1.80	-28.99*	1.66	-31.11*	1.78	-31.07*	1.72	-31.23*	1.75
Hours	-0.74	0.83	2.05	1.98	1.55	1.82	-2.01	1.24	-4.18	2.52	-2.11	3.87	-1.20	1.03
Age	-16.36*	2.55	-16.05*	2.55	-15.74*	2.58	-17.04*	2.56	-16.03*	2.54	-16.28*	2.56	-16.79*	2.59
Time x Hours	-1.26*	0.41	-1.13	1.06	0.76	1.28	-2.59*	0.61	-0.15	1.72	-4.62	2.60	-0.31	0.53
Time x Age	9.77*	1.19	9.63*	1.23	9.5*	1.27	8.82*	1.15	9.56*	1.26	9.58*	1.21	9.80*	1.25
Hours x Age	-0.48	0.64	-0.73	1.52	-2.0	1.16	1.27	0.99	-0.85	1.84	-2.11	2.77	0.44	0.78
Time x Hours x Age	0.86*	0.31	0.78	0.76	-0.88	0.87	2.16*	0.49	0.12	1.25	2.80	1.97	0.46	0.44

Note: β = beta coefficient; SE = standard error; * Denotes a statistically significant effect, $p < 0.05$. Full details of each model are presented in the supplemental appendix. Time was centered on the last year (2018) and scaled down by a factor of ten. Age was grand-mean centered on participants' mean age in 2018 (15.82). Hours was centered on the grand-mean average hours per year (in hundreds of hours) for each practice type. As such, intercepts can be interpreted as the estimated National Technical Points for a ski racer of the average age, who spent the average amount of time in practice, in 2018. (Marginal r^2 s: Total Hours = 0.385; Competition = 0.381; One-on-One = 0.376; Coach-Group = 0.388; Individual = 0.386; Play = 0.380; Indirect = 0.384).

but the effect of this extra time in practice was dependent on age, with younger ski racers demonstrating a faster rate of improvement. A similar pattern of effects was documented for coach-group practice, where there was no main-effect of Hours ($\beta = -2.013, p = 0.107$), but there were significant interactions of Time x Hours ($\beta = -2.59, p < 0.001$) and Time x Hours x Age ($\beta = 2.16, p < 0.001$). These interactions revealed that for both total hours of ski-related activities (not including indirect hours) and coach-led group practice, more time spent in practice each year was associated with greater improvement in technical rankings for young ski racers. Conversely, increased practice time was associated with smaller increases in ranking for older ski racers. Older individuals tended to have lower intercepts and shallower trajectories overall, so these interactions likely reflect the fact that it is harder to move from 20 to 10 than it is to move from 120 to 110 points in the national rankings.

Performance as a Function of Milestones. National technical points were regressed onto the effects of Time, Age in 2018, and the Age of Engagement variable that coded participants who first skied at, or before, 2 years of age versus those who started skiing later. As reflected in Fig. 3A, there were significant effects of Time ($\beta = -30.07, p < 0.001$), Age of Engagement ($\beta = -15.40, p = 0.024$), and Age in 2018 ($\beta = -15.39, p < 0.001$). Furthermore, there was a Time x Age in 2018 interaction ($\beta = 10.43, p < 0.001$). These effects reflect that rankings in technical events generally improved over time, but the rate of improvement was higher for younger ski racers. Furthermore, the effect of Age of Engagement indicated that ski racers who first started skiing earlier (≤ 2 yrs old) tended to have better rankings after controlling for the other factors in the model.

An Age of Competition variable coded participants who first competed at, or before, 7 years old (the median age for the sample) versus those who started competing later. There were significant effects of Time ($\beta = -28.18, p < 0.001$), and Age in 2018 ($\beta = -16.58, p < 0.001$), but no main effect of Age of Competition ($\beta = -7.73, p = 0.259$). Furthermore, there was a Time x Age in 2018 interaction ($\beta = 7.33, p < 0.001$). The rate of improvement in ranking generally improved over time and was greater for younger ski racers in the sample, but the age individuals began competing had no effect on performance trends (see

Fig. 3B).

We use an Age of Non-Ski Training variable to code participants who first started non-ski training at, or before, 12 years old (the median age) versus those who started non-ski training later. There was no main effect of Age of Non-Ski Training ($\beta = 0.85, p = 0.911$) however a significant Time x Age of Non-Ski Training interaction ($\beta = 11.51, p = 0.001$) was documented (see Fig. 3C), such that those who started non-ski training later had a flatter rate of improvement compared to those who started earlier.

An Age of Supervision variable coded participants who first started supervised ski training (i.e., with a coach) at, or before, 6 years old (the median) versus those who began supervised training later in development. There were significant main effects of Time ($\beta = -27.79, p < 0.001$) and Age in 2018 ($\beta = -14.93, p < 0.001$), but no main effect of Age of Supervision ($\beta = 4.50, p = 0.52$). There was a significant Time x Age in 2018 interaction ($\beta = 7.06, p < 0.001$), but no Time x Age of Supervision interaction ($\beta = -5.197, p < 0.157$). These results suggest that ski racers generally improve performance over time but younger athletes do not necessarily improve at a faster rate (see Fig. 3D).

Similarly, national technical points were regressed onto practice Time, Age in 2018, and the variable Other Sports, which coded participants who played $\geq 3,400$ h in other sports across development (the median) versus $< 3,400$ h. There were significant main effects of Time ($\beta = -26.80, p < 0.001$), Age in 2018 ($\beta = -14.52, p < 0.001$) and a Time x Age interaction ($\beta = 7.83, p < 0.001$). These results suggest that rank improved each year, with the effect being most pronounced in younger ski racers. Although there were no significant effects of Other Sports ($\beta = -11.02, p = 0.11$), Other Sports x Age ($\beta = -1.60, p = 0.76$), or Time x Other Sports x Age ($\beta = 3.31, p = 0.21$), a significant Time x Other Sports interaction was documented ($\beta = -8.87, p = 0.014$). This interaction indicated that ski racers who spent more time playing other sports later in their development improved their ranking at a faster rate when compared to those who spent less time in other sports.

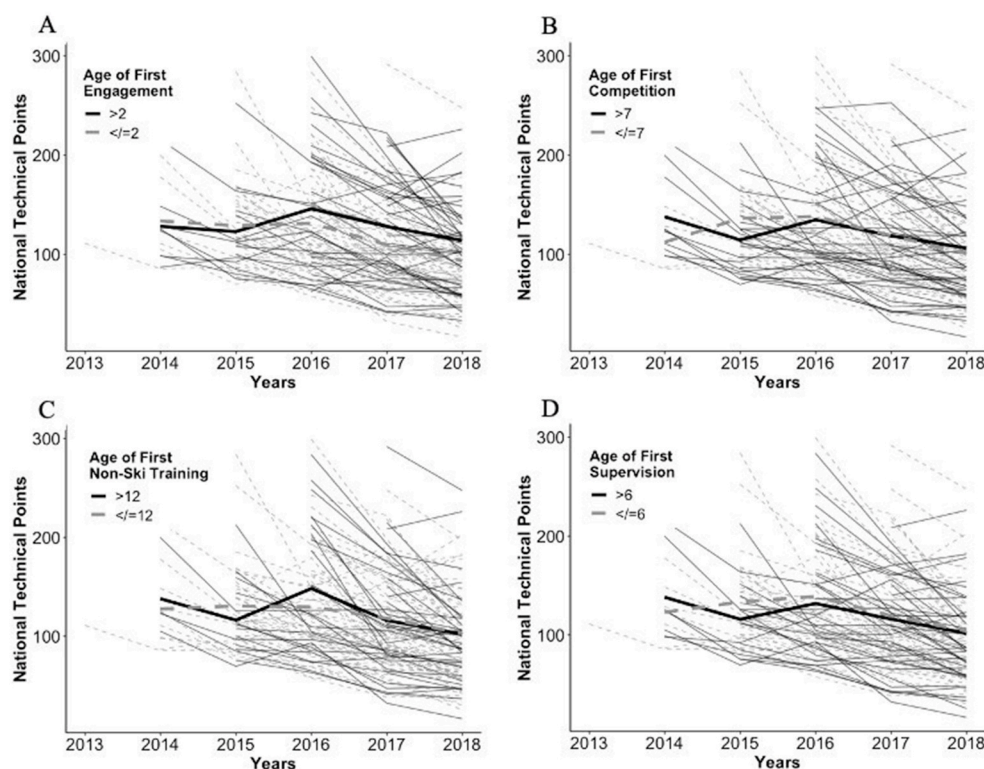


Fig. 3. (A) National technical points as a function of time and age of first engagement (Marginal $r^2 = 0.384$). (B) National technical points as a function of time and age of first competition (Marginal $r^2 = 0.373$). (C) National technical points as a function of time and age of first non-ski training (Marginal $r^2 = 0.383$). (D) National technical points as a function of time and age of first supervision (Marginal $r^2 = 0.378$). Individual participant data are shown as thin lines. The aggregate data are shown as thick lines, averaging across participants at each time point (year). Line color and type varies as described in the individual figure legends.

4. Discussion

In the present study, we examined the relative importance of engagement in different types of practice activities for superior performance in alpine ski racing, as well as contributions from both the specialization and diversification developmental pathways. We extend previous work in the sport expertise literature by examining these developmental pathways over time, finding that athletes adopt a multifaceted developmental approach to improve performance in alpine ski racing. We used linear models to analyze developmental trajectories and identify specific practice activity trends and performance across time. Specifically, we used mixed-effects regression analyses to model changes in alpine ski racing practice and objective performance markers over time based on the relative influence of sport-specific milestones, hours of engagement, and types of practice as assessed through retrospective participation history profiling (Ford, Williams, & Yates, 2010). Consequently, this approach helps to assimilate a better understanding of the prototypical pathway to excellence in alpine ski racing within U.S. High Performance Centers, with potential implications for other sports.

The current sample of alpine ski racers demonstrated trends towards early engagement, as revealed by the average age of first engagement in the sport of skiing ($M = 2.71 \pm 1.53$ yrs), age of first supervised ski training ($M = 6.17 \pm 2.02$ yrs), age of first ski racing competition ($M = 7.45 \pm 1.75$ yrs), and age of first non-ski training ($M = 11.02 \pm 2.54$ yrs), all occurring before adolescence. In support of the early engagement hypothesis, these milestones occur approximately two years earlier than previously reported milestones in other sporting domains such as soccer (Ford, Williams, & Hodges, 2009; Roca et al., 2012), but are similar to sports like gymnastics (Beamer, Côté, & Ericsson, 1999; Law et al., 2007). Anecdotally, alpine ski racing has heightened physical, tactical, and psychological demands compared to some domains due to environmental and task constraints (Fawver et al., 2019; Gilgien, Reid, Raschner, Supej, & Holmberg, 2018); therefore, earlier engagement allows greater volumes of exposure and experience to deal with the potential adversity associated with alpine ski racing at elite competitive levels (Ford, Williams, & Hodges, 2009). Alternatively, the early age of engagement may merely be indicative of the fact that these athletes resided in close proximity to ski areas or came from families with the financial means and interest to support the activity.

Our data indicated that sub-elite level ski racers are training close to 500 total hours a year, which equates to an average of 25 h per week across a ski season (~20 weeks of the year). According to Gilgien et al. (2018), Olympic-level alpine ski racers train and compete for, on average, 130–150 days of the calendar year. Although the distribution of training hours may vary, these adolescent athletes are matching the yearly training load for Olympic-level skiers indicated by Gilgien et al. (2018), and our mixed-effects models reveal that skiers accumulate the majority of these hours via coach-led group practice, competition, and indirect exposure time. The high volume of coach-led group hours is primarily indicative of the ski academy structure, which groups athletes by age and gender (e.g., women U16's) for training purposes throughout the school year. Similarly, academies foster a competitive environment, entering athletes regularly into regional, national, and cup events, which may explain the consistently high volume of hours in competition for this specific developmental population.

From first engagement (~2–3 years old) up until 18 years of age, the average alpine ski racer in our sample is engaging in nearly 9,000 h of practice in skiing ($M = 8,577$ total h). As expected, hours models estimated that engagement in practice hours increases over time and that this rate increases in later years. Following this positive association, time spent in coach-led group, individual, indirect, and coach one-on-one practice all increased over time. Our models further suggest that increasing the volume of sport-specific practice above the average allows ski racers to enhance their skill level and improve performance in adolescence. This finding supports previous published reports that have highlighted the positive relationship between greater practice hours and

improved performance (Baker et al., 2005; Ford & Williams, 2017; Helsen et al., 1998; Hodges et al., 2004; Law et al., 2007; Starkes et al., 1996; Williams & Ford, 2008). Furthermore, decreases in hours of play in later years suggests that ski racers adapt practice over time to be more training-focused, with the aim of improving sport-specific skills and overall performance. The curvilinear effect of increased total practice hours over time may reflect the increasing competitive status of athletes. For example, training hours intensified from 562 total practice hours on average at 13 years old, to 710 total hours at 14 years old, which may be indicative of the introduction to competition on the national circuit and/or the stage at which athletes begin to intensify and accumulate greater hours in a primary sport (Moesch et al., 2011).

Although trends indicate a positive association between earlier first engagement and increased practice hours, this relationship did not reach the threshold for statistical significance. The age of first engagement objectively states an initial direct exposure to skiing, yet the milestone itself is not necessarily indicative of the child taking up skiing as a serious sport, engaging in regular practice, or accumulating coach-led or individual training hours. The age of first engaging in supervised practice, on the other hand, may better reflect an athletes' intent to continue participation in the sport for developing expertise (Baker et al., 2005; Ericsson et al., 1993). We found that earlier engagement in supervised practice, competition, and non-ski specific training were each associated with greater total practice hours and steeper trajectories for accumulated hours per year. The early engagement hypothesis, suggests that engaging early allows more time to accumulate more hours in training and competition over athletes who engage later (Ford, Williams, & Hodges, 2009). These early engagement and performance trends seen in this study align well with the predictions of the early engagement hypothesis (Ford, Williams, & Hodges, 2009). Namely, early engagement may allow for a greater accumulation of structured coach-led practice, but not necessarily structured individual practice. Additionally, early engagement in alpine ski racing indicates increased hours of play, which according to Ford, Williams, and Hodges (2009) is an important variable to differentiate skill levels. Although some support is provided for the Deliberate Practice Framework, such as the relationships between increased structured practice, earlier engagement, and improved performance, it is clear that these results present a better fit with the predictions and intricacies of the early engagement hypothesis (Ford, Williams, & Hodges, 2009).

Ski racers in this academy setting typically continue to engage in high volumes of hours in others sports throughout development, especially for younger athletes, indicating that while alpine ski racers exhibit some characteristics associated with specialization (i.e., accumulating considerable practice time in ski racing), they maintain aspects of diversification (i.e., ~5 other sports played before the age of 18). Additionally, ski racers attending professional development academies engage in a high volume of hours in seasonal school sports, such as lacrosse and soccer. This latter finding contradicts previous published reports that suggest that the number of sports played is reduced when athletes reach adolescence and begin to prioritize practice in a single primary sport (Moesch et al., 2011). It appears that sub-elite ski racers in the US are typically involved in many curriculum-based sports and school teams, thus this finding may not represent skiing specifically, but academy-structured sport programs in general. Although the majority of elite alpine ski racers come through the academy system, this may be a misrepresentation of the population as a whole. It would be interesting to see if involvement in other sports decreases without a structured academy curriculum that includes physical education and school sports teams, how this participation changes after athletes reach 18 years old and/or progress to the Olympic level, and if greater diversification becomes predictive of success for these adolescent skiers when they reach adulthood, as proposed elsewhere (e.g., DiFiori et al., 2018). Alternatively, the high level of diversification seen in these ski racers may arise from distinctions in ski activity between “in-season” and “off-season” as well as the structure of the ski academy. During the summer months,

many academy ski racers attend training camps in Southern Hemisphere countries experiencing the winter season (i.e., New Zealand). Usually, these camps last for around a month, so it is possible that high diversification occurs when other summer months are occupied by sports other than skiing. Clearly, the seasonal constraints of alpine ski racing allow ample opportunity for academy ski racers to engage in other sports activities during the notional off-season.

As hypothesized, we report a decreased rate of improvement in rankings as ski racers get older. Initially, ranks improve at a faster rate due to the system algorithm (see supplemental appendix for a full description), but as individuals get older, they have typically competed for a number of years and in closer-seeded races that slows their overall rate of improvement. Additionally, improvement rates may plateau as older athletes are exposed to 'ceiling effects.' As expected, we found that performance improved each year, but when examining changes in performance trajectory as a function of time in practice, our models revealed that rate of improvement depended on age. Specifically, interactions between time, age, and total hours further supported the notion that engaging in greater than average total hours improved rank, but that younger athletes tend to improve at a faster rate. Traditional models of sport development suggest that in order to develop expertise, athletes must accumulate a greater number of deliberate practice hours (Ericsson et al., 1993). However, our findings, and more recent criticism of the overemphasis on deliberate practice, suggest that it is not only the volume of hours, but also the type (e.g., both sport-specific and non-sport-specific) and content of practice (e.g., unstructured play or structured coach-led practice) that best predicts performance (Coutinho, Mesquita, & Fonseca, 2016). Linear regression modeling allowed tracking of different types of sport engagement over time (i.e., coach-led group, coach-led one-to-one, individual, indirect, play, competition hours). Findings indicated that coach-led group practice is the most dominant practice type within the ski academy structure (average hours per year = 168.5 ± 155.65). No significant main effect of coach-led group hours was documented, but similar to total hours, higher-level interactions revealed that the more time spent in coach-led practice was associated with a faster rate of ranking improvement for younger athletes. Faster rate of ranking improvement for older skiers was associated with more time in other sports when compared to those who spent less time in other sports, suggesting diversification might become more predictive as athletes get older (e.g., DiFiori et al., 2018) and demonstrating the need for further research with adult skiers. We report no main effects of hours or higher-order interactions, including hours for other practice types. The absence of other effects suggests that coach-led group practice is the most influential practice type associated with performance improvements in early adolescence. This is unsurprising, as practice activities such as play and individual practice had the lowest average hours of engagement in a given year. Most likely, the academy system limits time on the slopes for individual training, or recreational skiing, or that athletes are too tired to engage in further exposure on the slopes due to injury risk. In ski racing specifically, it appears that engaging in relatively more play and individual practice may not be necessary for developing expertise (Kowal & Fortier, 1999).

A main effect of Age of First Engagement suggests that athletes who tend to engage in the sport earlier also tend to have improved performance rankings. Although age of first engagement may only refer to a one-time sporting exposure, in some cases this milestone marks the beginning of development and thus is key for any athlete to highlight interest and engagement at a younger age. A significant interaction revealed that ski racers who engage in non-ski training earlier improve at a faster rate, with this effect being pronounced in younger individuals. This finding potentially highlights the advantages of general fitness, such as weight training, in order to manage the physical demands and injury risk in the domain. Although we did not account for measurements of injury in this study, such training may ameliorate injury risk through reduced overtraining and increased conditioning (Gustafsson, Sagar, & Stenling, 2017; McGuine et al., 2017; Moen, Myhre, Sandbakk,

& Moen, 2016).

An earlier engagement in competition and earlier engagement in supervised practice was associated with improved performance. The process of engaging earlier in competition may help develop essential psychological attributes, as a result of greater competitive experience and opportunities to combat adversity and/or challenge during a race. This suggestion is supported by Fawver et al. (2019) who report that skiers who engaged earlier are grittier and hold higher personal standards of perfection, which the authors also reported to be associated with improved rank. The USSS national circuit that ski racers enter at 14 years of age may be a huge step up from the youth competition, and more reliably represents the level of competition needed to elevate performance through competitive exposure. Furthermore, the effects of earlier supervised skiing may not be readily evident, as typically the athletes who have made it onto the national circuit to compete are already within the academy system.

Mixed-effects models indicated that ski racers are engaging in a high volume of hours in multiple sports across development. Specifically, we report a significant time and other sports interaction when analyzing performance as a function of hours in other sports. This means athletes who maintained engagement in other sports across development improved their rank at a faster rate later on when older compared to those who spent less time involved in other sports (e.g., DiFiori et al., 2018). It may be that because specialists increased their ranking faster earlier, diversifiers were increasing later across the lower rankings, which is easier to do than increasing across higher rankings. The engagement in other sports may act as a channel to acquire or transfer skills to ski racing through other domains. Alternatively, other sports could be useful to maintain general fitness during the summer months when ski racers are away from regular ski-season training and overseas training camps. These athletes are likely more physically fit and athletically inclined (DeCouto, Taylor, & Williams, 2021). Thus, engagement in other sports could potentially reduce physical and psychological risk associated with high volumes of prolonging and demanding training, while potentially improving performance (Fabricant et al., 2016; Kaleth & Mikesky, 2010).

In summary, we examined the developmental trajectories of sub-elite level adolescent alpine ski racers using linear mixed-effects regression analysis techniques. This novel approach yielded evidence for the non-dichotomous nature of specialization and diversification as distinct talent development pathways. Moreover, we detailed the various engagement activities, practice types, and practice volume that are associated with the steepest developmental performance trajectory in alpine ski racers. It appears that athletes in this domain engage in the sport from an early age and accumulate a high volume of ski-specific hours, with the majority of these hours being in supervised, coach-led practice. Overall, based on athletes also participating in a variety of other sports during development, findings suggest that alpine ski racers follow a multifaceted developmental continuum, with influence from both the specialization and diversification pathways. As a result, the physically and psychologically demanding nature of alpine ski racing could be modulated by a pathway that involves diverse sporting engagement. Overall, our data highlight the value of using mixed-effects regression to model developmental trajectories over time across various professional domains and provide support for the early engagement hypothesis.

CRedit authorship contribution statement

Rhiannon L. Cowan: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Validation, Visualization, Writing – original draft, Writing – review & editing. **Bradley Fawver:** Conceptualization, Data curation, Investigation, Methodology, Project administration, Writing – review & editing. **Keith R. Lohse:** Conceptualization, Formal analysis, Investigation, Methodology, Software, Supervision, Writing – review & editing. **Troy Taylor:**

Funding acquisition, Project administration. **Paul R. Ford:** Conceptualization, Writing – review & editing. **A. Mark Williams:** Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

Funding for this study was provided by the United States Olympic Committee (USOC) through United States Ski and Snowboard (USSS) and the University of Utah Sports Medicine and Sports Science Institute. Contacts at USSS had little involvement in the measures and study design and no involvement in the analysis, interpretation, dissemination of findings, or submission decision. There are no other conflicts of interest to declare.

Acknowledgements

We would like to thank Sarah Taylor for her assistance with data reduction and initial analysis.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.psychsport.2021.101942>.

References

- Abbott, A., & Collins, D. (2004). Eliminating the dichotomy between theory and practice in talent identification and development: Considering the role of psychology. *Journal of Sports Sciences*, 22(5), 395–408. <https://doi.org/10.1080/02640410410001675324>
- Abernethy, B., Baker, J., & Côté, J. (2005). Transfer of pattern recall skills may contribute to the development of sport expertise. *Applied Cognitive Psychology*, 19(6), 705–718. <https://doi.org/10.1002/acp.1102>
- Baker, J., Cobley, S., & Fraser-Thomas, J. (2009). What do we know about early sport specialization? Not much! *High Ability Studies*, 20(1), 77–89. <https://doi.org/10.1080/13598130902860507>
- Baker, J., Côté, J., & Deakin, J. (2005). Expertise in ultra-endurance triathletes early sport involvement, training structure, and the theory of deliberate practice. *Journal of Applied Sport Psychology*, 17(1), 64–78.
- Baker, J., & Young, B. (2014). 20 years later: Deliberate practice and the development of expertise in sport. *International Review of Sport and Exercise Psychology*, 7(1), 135–157. <https://doi.org/10.1080/1750984X.2014.896024>
- Baker, J. (2003). Early specialization in youth sport: a requirement for adult expertise? *High Ability Studies*, 14(1), 85–94.
- Bates, D., Mächler, M., Zurich, E., Bolker, B. M., & Walker, S. C. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1), 1–51.
- Beamer, M., Côté, J., & Ericsson, K. A. (1999). A comparison between international and provincial level gymnasts in their pursuit of sport expertise. *Proceedings of the 10th European Congress of Sport Psychology*, 90–92.
- Byrnes, J. (2008). *Re: Coefficient of determination (R²) when Using lme*.
- Cote, J., Baker, J., & Abernethy, B. (2003). From play to practice: A developmental framework for the acquisition of expertise in team sports. In J. Starkes, & K. A. Ericsson (Eds.), *Expert performance in sports: Advances in research on sport expertise*.
- Côté, J., Baker, J., & Abernethy, B. (2007). Practice and play in the development of sport expertise. In G. Tenenbaum, & R. C. Eklund (Eds.), *Handbook of sport psychology*. Wiley.
- Côté, J., Lidor, R., & Hackfort, D. (2009). ISSP position stand: To sample or to specialize? Seven postulates about youth sport activities that lead to continued participation and elite performance. *International Journal of Sport and Exercise Psychology*, 7(1), 7–17. <https://doi.org/10.1080/1612197X.2009.967189>
- Côté, J., & Verma, M. (2014). The developmental model of sport participation: 15 years after its first conceptualization. *Science & Sports*, 29, S63–S69. <https://doi.org/10.1016/j.scispo.2014.08.133>
- Coutinho, P., Mesquita, I., Davids, K., Fonseca, A. M., & Côté, J. (2016). How structured and unstructured sport activities aid the development of expertise in volleyball players. *Psychology of Sport and Exercise*, 25, 51–59.
- Coutinho, P., Mesquita, I., & Fonseca, A. M. (2016). Talent development in sport: A critical review of pathways to expert performance. *International Journal of Sports Science & Coaching*, 11(2), 279–293. <https://doi.org/10.1177/1747954116637499>
- DeCouto, B. S., Taylor, T., & Williams, A. M. (2021). Physical fitness is associated with better technical performance in adolescent alpine ski racers after controlling for practice time: A retrospective regression analysis. *Journal of Sports Sciences*, 39(4), 380–387.
- DiFiore, J. P., Güllich, A., Brenner, J. S., Côté, J., Hainline, B., Ryan, E., & Malina, R. M. (2018). The NBA and youth basketball: recommendations for promoting a healthy and positive experience. *Sports Medicine*, 48(9), 2053–2065.
- Ericsson, K. A. (2006). The influence of experience and deliberate practice on the development of superior expert performance. In K. A. Ericsson, N. Charness, P. Feltovich, & R. Hoffman (Eds.), *Cambridge handbook of expertise and expert performance*. Cambridge, UK: Cambridge University Press.
- Ericsson, K. A. (2020). Towards a science of the acquisition of expert performance in sports: Clarifying the differences between deliberate practice and other types of practice. *Journal of Sports Sciences*, 38(2), 159–176. <https://doi.org/10.1080/02640414.2019.1688618>
- Ericsson, K. A., & Harwell, K. (2019). Deliberate practice and proposed limits on the effects of practice on the acquisition of expert performance: Why the original definition matters and recommendations for future research. *Frontiers in Psychology*, 10(2396).
- Ericsson, K. A., Krampe, R. T., & Tesch-Römer, C. (1993). The role of deliberate practice in the acquisition of expert performance. *Psychological Review*, 100(3), 363–406. <https://doi.org/10.1037/0033-295X.100.3.363>
- Ericsson, K. A., & Smith, J. (1991). *Toward a general theory of expertise: Prospects and limits*. New York, NY: Cambridge University Press.
- Ericsson, K. A., Whyte, J., Ward, P., Goulding, J., & Williams, A. M. (2017). Expert performance in nursing: Reviewing research on expertise in nursing within the framework of the expert-performance approach. *ANS. Advances in Nursing Science*, 30(1), E58–E71. <https://doi.org/10.1097/00012272-200701000-00014>
- Fabricant, P. D., Lakomkin, N., Sugimoto, D., Tepolt, F. A., Straccioli, A., Kocher, M. S., et al. (2016). Youth sports specialization and musculoskeletal injury: a systematic review of the literature. *The Physician and sportsmedicine*, 44(3), 257–262.
- Fawver, B., Cowan, R. L., DeCouto, B. S., Lohse, K. R., Podlog, L., & Williams, A. M. (2020). Psychological characteristics, sport engagement, and performance in alpine skiers. *Psychology of Sport and Exercise*, 47, Article 101616. Pre-print(November 2019), 101616. <https://doi.org/10.1016/j.psychsport.2019.101616>
- Ford, P. R., Low, J., McRobert, A. P., & Williams, A. M. (2010a). Developmental activities that contribute to high or Low performance by elite cricket batters when recognizing type of delivery from bowlers' advanced postural cues. *Journal of Sport & Exercise Psychology*, 32(5), 638–654. <https://doi.org/10.1123/jsep.32.5.638>
- Ford, P. R., Ward, P., Hodges, N. J., & Mark Williams, A. (2009a). The role of deliberate practice and play in career progression in sport: The early engagement hypothesis. *High Ability Studies*, 20(1), 65–75. <https://doi.org/10.1080/13598130902860721>
- Ford, P. R., & Williams, A. M. (2012). The developmental activities engaged in by elite youth soccer players who progressed to professional status compared to those who did not. *Psychology of Sport and Exercise*, 13. <https://doi.org/10.1016/j.psychsport.2011.09.004>
- Ford, P. R., & Williams, M. A. (2017). *Early specialization and diversification. In identification and development in sport*. Taylor & Francis.
- Ford, P. R., Williams, M. A., & Hodges, N. J. (2009b). *The role of deliberate practice and play in career progression in sport: The early engagement hypothesis*.
- Ford, P. R., Williams, A. M., & Yates, I. (2010b). Towards enhanced methods of examining the practice activities and instructional behaviours of sport coaches. *Journal of Sports Sciences*, 28(14), 1627–1630. <https://doi.org/10.1080/02640414.2010.522590>
- Garcia, T. P., & Marder, K. (2017). Statistical approaches to longitudinal data analysis in neurodegenerative diseases: Huntington's disease as a model. In *Current neurology and neuroscience reports* (Vol. 1). Current Medicine Group LLC. <https://doi.org/10.1007/s11910-017-0723-4>
- Gilgen, M., Reid, R., Raschner, C., Supej, M., & Holmberg, H.-C. (2018). The training of olympic alpine ski racers. *Frontiers in physiology*. Retrieved from <https://www.frontiersin.org/article/10.3389/fphys.2018.01772>
- Gustafsson, H., Sagar, S. S., & Stenling, A. (2017). Fear of failure, psychological stress, and burnout among adolescent athletes competing in high level sport. *Scandinavian Journal of Medicine & Science in Sports*, 27(12), 2091–2102.
- Helsen, W. F., Starkes, J. L., & Hodges, N. J. (1998). Team sports and the theory of deliberate practice. *Journal of Sport & Exercise Psychology*, 20(1), 12–34.
- Hodges, N. J., Starkes, J. L., Nananidou, A., Kerr, T., & Weir, P. L. (2004). Predicting performance times from deliberate practice hours for triathletes and swimmers: What, when, and where is practice important? *Journal of Experimental Psychology: Applied*, 10(4), 219–237. <https://doi.org/10.1037/1076-898X.10.4.219>
- Jayanthi, N., Pinkham, C., Dugas, L., Patrick, B., & LaBella, C. (2013). Sports specialization in young athletes: Evidence-based recommendations. *Sport Health*, 5(3), 251–257. <https://doi.org/10.1177/1941738112464626>
- Kaleth, A. S., & Mikesky, A. E. (2010). Impact of early sport specialization: A physiological perspective. *Journal of Physical Education, Recreation & Dance*, 81(8), 29–37.
- Kowal, J., & Fortier, M. S. (1999). Motivational determinants of flow: Contributions from self-determination theory. *The Journal of Social Psychology*, 139(3), 355–368.
- Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. B. (2017). lmerTest package: Tests in linear mixed effects models. *Journal of Statistical Software*, 82(13). <https://doi.org/10.18637/jss.v082.i13>
- Law, M. P., Côté, J., & Ericsson, K. A. (2007). Characteristics of expert development in rhythmic gymnastics: A retrospective study. *International Journal of Sport and Exercise Psychology*, 5(1), 82–103. <https://doi.org/10.1080/1612197X.2008.967184>
- Long, J. D. (2012). *Longitudinal data analysis for the behavioral sciences using R*. Sage Publications.
- Lüdtke, D. (2020). *sjstats: Statistical functions for regression models* Version 0.18.0. 10.5281/zenodo.1284472.
- Macnamara, B. N., Hambrick, D. Z., & Oswald, F. L. (2014). Deliberate practice and performance in music, games, sports, education, and professions: A meta-analysis. *Psychological Science*, 25(8), 1608–1618.

- McGuine, T. A., Post, E. G., Hetzel, S. J., Brooks, M. A., Triggsted, S., & Bell, D. R. (2017). A prospective study on the effect of sport specialization on lower extremity injury rates in high school athletes. *The American Journal of Sports Medicine*, 45(12), 2706–2712.
- Moesch, K., Elbe, A. M., Hauge, M. L. T., & Wikman, J. M. (2011). Late specialization: The key to success in centimeters, grams, or seconds (cgs) sports. *Scandinavian Journal of Medicine & Science in Sports*, 21(6), 1–9. <https://doi.org/10.1111/j.1600-0838.2010.01280.x>
- Moen, F., Myhre, K., Sandbakk, Ø., & Moen, F. (2016). Psychological determinants of burnout, illness and injury among elite junior athletes. *The Sport Journal*, 19, 1–14.
- O'Connor, D., Larkin, P., & Williams, A. M. (2016). Talent identification and selection in elite youth football: An Australian context. *European Journal of Sport Science*. <https://doi.org/10.1080/17461391.2016.1151945>
- Orr, B. (2007). Cross-country skiing USSA points as a predictor of future performance among junior skiers. *The Sport Journal*, 10(4), 16–25. <http://ezproxy.library.ubc.ca/login?url=http://search.ebscohost.com/login.aspx?direct=true&db=sph&AN=27230762&login.asp&site=ehost-live&scope=site>.
- R Core Team. (2017). R: A language and environment for statistical computing. *R Foundation for Statistical Computing*. <https://www.R-project.org/>.
- Roca, A., Williams, A. M., & Ford, P. R. (2012). Developmental activities and the acquisition of superior anticipation and decision making in soccer players. *Journal of Sports Sciences*, 30(15), 1643–1652. <https://doi.org/10.1080/02640414.2012.701761>
- Starkes, J. L., Deakin, J., Allard, F., Hodges, N. J., & Hayes, A. (1996). Deliberate practice in sports: What is it anyway? In K. A. Ericsson (Ed.), *The road to excellence: The acquisition of expert performance in the arts and sciences, sports, and games*. Mahwah, NJ: Erlbaum.
- Wall, M., & Côté, J. (2007). Developmental activities that lead to dropout and investment in sport. *Physical Education and Sport Pedagogy*, 12(1), 77–87.
- Walters, B. K., Read, C. R., & Estes, A. R. (2018). The effects of resistance training, overtraining, and early specialization on youth athlete injury and development. *The Journal of Sports Medicine and Physical Fitness*, 58(9), 1339–1348.
- Ward, P., Hodges, N. J., Starkes, J. L., & Williams, M. A. (2007). The road to excellence: Deliberate practice and the development of expertise. *High Ability Studies*, 18(2), 119–153. <https://doi.org/10.1080/13598130701709715>
- Williams, A. M., & Ford, P. R. (2008). Expertise and expert performance in sport. *International Review of Sport and Exercise Psychology*, 1(1), 4–18.